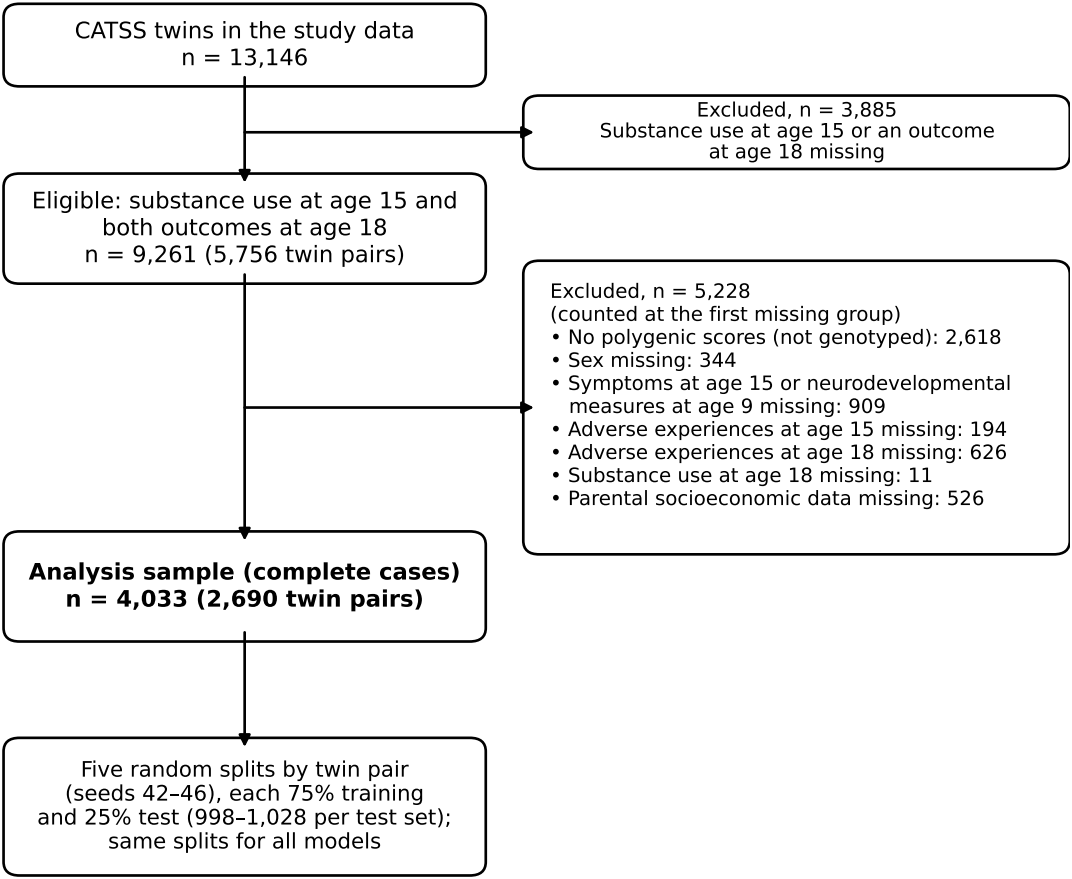
CATSS twins in the study data
n = 13,146
Excluded, n = 3,885
Substance use at age 15 or an outcome
at age 18 missing
Eligible: substance use at age 15 and
both outcomes at age 18
n = 9,261 (5,756 twin pairs)
Excluded, n = 5,228
(counted at the first missing group)
• No polygenic scores (not genotyped): 2,618
• Sex missing: 344
• Symptoms at age 15 or neurodevelopmental
measures at age 9 missing: 909
• Adverse experiences at age 15 missing: 194
• Adverse experiences at age 18 missing: 626
• Substance use at age 18 missing: 11
• Parental socioeconomic data missing: 526
Analysis sample (complete cases)
n = 4,033 (2,690 twin pairs)
Five random splits by twin pair
(seeds 42–46), each 75% training
and 25% test (998–1,028 per test set);
same splits for all models